PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT JOINT EXAMINATION
FORM ONE

031/1 PHYSICS

TIME: 2:00 HRS APRIL 2019

MARKING SCHEME

SECTION A (30 MARKS)

1. 1@=(10marks)

i	Ii	iii	iv	v	vi	vii	viii	ix	x
B	C	A	A	B	C	A	B	B	C

2. 2@= (10marks)

LIST A	i	ii	iii	iv	V
LIST B	E	C	G	D	H

3. (i) Thermometer
   (ii) Flammable
   (iii) Hypothesis **2@=(10marks)**
   (iv) Time
   (v) Volume
4. (a) Physics is the study of the relationship between matter and energy (**2marks)**

   (b) Four importance of studying physics are; **2@=(8marks)**
   i. It imparts knowledge and skills on how to make things
   ii. It helps people to acquire different carriers or professions
   iii. It is a fun since it brings entertainments to the people studying it
   iv. It helps people to understand the physical properties of matter
5. (a) Measurement is the process of assigning numbers to various observations or events

   (b) Differences between mass and weight

mass	weight
i. It is the quantity of matter it contains	-It is the force of attraction exerted by a body on earth
ii. Its SI-unit is kilogram (Kg)	-Its SI-unit is Newton (N) It is measured by spring balance
iii. It is measured by beam balance	-It changes with position
iv. It does not change with position	It is a vector quantity
v. It is scalar quantity	It is derived quantity
vi. It is fundamental basic quantity	

(c) (i) Mass (ii) Time (iii) Length

6. (a) i. Laboratory is a special room or building designed for scientific experiments
ii. Energy is the ability of doing work**2@ =4marks**
(b) Six laboratory rules are
i. Do not enter in the laboratory without permission of your teacher or laboratory assistance
ii.Do not do any experiment unless your teacher permits you to do so **1@= (6marks)**
iii. Do not start an experiment before you get information about the proper procedure to be Followed
iv. Do not use broken apparatus
v. Avoid tasting, eating or drinking anything in the laboratory
vi. Report all accidents and injuries to the teacher or laboratory assistant.

7. (a) First Aid is an immediate help given to a victim or injured person before sending him/her to the Hospital for further treatment (**3marks)**
(b) Four importance of first Aid are;
i. It helps to preserve life
ii. It prevents the victim's condition from becoming worse**1@= (4marks)**
iii. It promotes recovery by bringing hope and encouragement to the victim
iv.It helps to reduce pain and suffering

(c) i. Micrometer screw gauge is used to measure diameter of a thin objects

ii. Measuring cylinder is used to measure volume of liquids**1@=(3marks)**

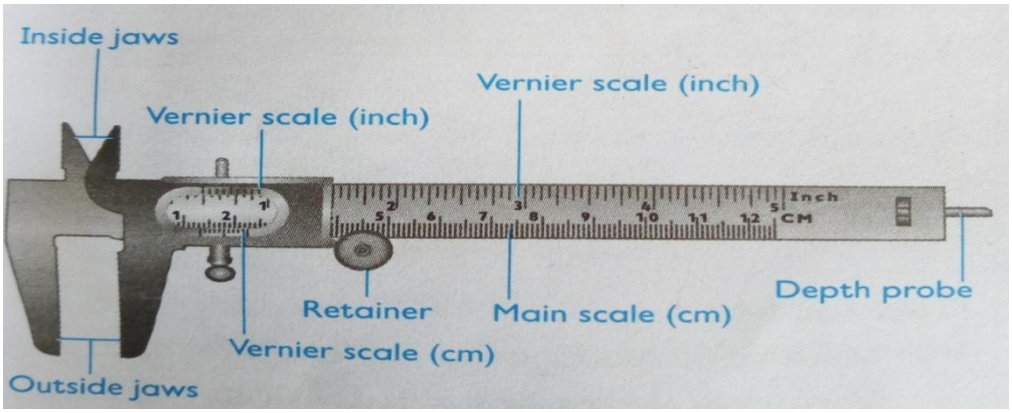
iii. Slotted masses are used for measuring the quantity of matter.

8. (a) Volume is the amount of space which is occupied by a substance**(1 mark)**
(b) Given
Mass of stone, m = 410g
Initial volume, $V_1 = 500cm^3$
Final volume, $V_2 = 700cm^3$
(i) **Volum**e of stone, $V = V_2 - V_1 = (700 - 500)cm^3 = 200\ cm^3$ **(3 marks)**
(ii) Density of stone, ρ= $\frac{mass\ of\ stone}{volume\ of\ stone} = \frac{410g}{200\ cm^3} = 2.05g/cm^3$**(3 marks)**
(iii) **R**elative density of stone, R.D $= \frac{Density\ of\ stone}{Density\ of\ water} = \frac{2.05g/cm^3}{1.0g/cm^3} = 2.05$ **(3 marks)**

9. (a) Scientific method is a set of techniques used by scientists to investigate a problem or answer question**(1mark)**
(b) (i) Problem identification (observation)
(ii) Asking questions(iii) Hypothesis

(iv) Experiment
(v) Data correction and analysis
(vi) Data interpretation
(vii) Data presentation
(viii) Drawing a conclusion

10. (a) Vernier caliper is the instrument used to measure the length to an accuracy of 0.01cm. **(2 marks)**
(b) Diagram of vernier caliper **(8 marks)**


Inside jaws
Vernier scale (inch)
Vernier scale (inch)
Inch
CM
Depth probe
Retainer
Main scale (cm)
Vernier scale (cm)
Outside jaws